

## Test 13 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

| #  | Benchmark               | Answer                                                                    | Explanation                                                                                                |
|----|-------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 1  | MA.6.DP.1.1<br>moderate | each student in the class                                                 | Asking about a whole group produces data that vary; asking about one person produces a single answer.      |
| 2  | MA.6.AR.1.4<br>moderate | $6(x + 8)$ , $6x + 48$ , $48 + 6x$                                        | Equivalent expressions have the same value for every value of $x$ .                                        |
| 3  | MA.6.NSO.1.3<br>easy    | 9, -9                                                                     | Both 9 and -9 are 9 units from 0.                                                                          |
| 4  | MA.6.DP.1.3<br>moderate | 13, 15, 18, 22                                                            | The box itself covers the values from the lower quartile to the upper quartile.                            |
| 5  | MA.6.NSO.3.1<br>easy    | 3                                                                         | The greatest number of identical bags is the greatest common factor.                                       |
| 6  | MA.6.AR.3.1<br>moderate | 4:7, 12:21, 16:28                                                         | Equivalent ratios simplify to the same pair of numbers in the same order.                                  |
| 7  | MA.6.AR.3.3<br>moderate | 5:6, 10:12, 20:24                                                         | Multiplying both parts of a ratio by the same number keeps it equivalent; adding the same number does not. |
| 8  | MA.6.AR.1.3<br>moderate | $2x + 10$ , $10 + 2x$ , $2(x) + 10$                                       | Substitute $x$ and evaluate each expression using the order of operations.                                 |
| 9  | MA.6.GR.2.3<br>moderate | $4 \times 5 \times 2$ , $20 \times 2$ , $2 \times 5 \times 4$             | Volume is the product of the three edge lengths, which is also the area of the base times the height.      |
| 10 | MA.6.GR.1.3<br>moderate | (3, -3), (3, 2), (3, 6)                                                   | Every point on a vertical line has the same $x$ -coordinate.                                               |
| 11 | MA.6.DP.1.6<br>moderate | Mean, Range, Maximum, Median                                              | Adding a large value shifts the total and the greatest value, but it may leave the least value unchanged.  |
| 12 | MA.6.AR.2.2<br>moderate | $x + 8 = 23$ , $x - 8 = 7$ , $8 + x = 23$                                 | Substitute 15 for $x$ and keep the equations that stay balanced.                                           |
| 13 | MA.6.GR.2.1<br>moderate | $(8 \times 9) \div 2$ , $0.5 \times 8 \times 9$ , $8 \times 9 \times 0.5$ | The area of a triangle is half the product of the base and the height.                                     |
| 14 | MA.6.NSO.1.2<br>easy    | -12, 12                                                                   | Below zero is negative, and the opposite of a negative number is positive.                                 |
| 15 | MA.6.AR.2.1<br>moderate | 6, 8, 9                                                                   | The symbol $\leq$ includes 9 itself.                                                                       |
| 16 | MA.6.AR.2.3<br>moderate | $7x = 42$ , $x \div 7 = 6/7$ , $42 = 7x$                                  | Substitute 6 for $x$ and keep the equations that are true.                                                 |
| 17 | MA.6.DP.1.5<br>moderate | 0-4, 10-14, 15-19                                                         | Count the values in each interval and compare.                                                             |

| #  | Benchmark                   | Answer                                                                                                      | Explanation                                                                                       |
|----|-----------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 18 | MA.6.NSO.3.2<br>easy        | 4                                                                                                           | Find the largest number that divides both addends.                                                |
| 19 | MA.6.DP.1.4<br>moderate     | 6, 7                                                                                                        | Look for the columns that have exactly 2 dots.                                                    |
| 20 | MA.6.NSO.3.4<br>easy        | $2 \times 5^2$                                                                                              | Divide by the smallest prime repeatedly until only primes remain.                                 |
| 21 | MA.6.NSO.2.3<br>easy        | -3                                                                                                          | Multiply to find the ribbon used, then subtract from the whole roll.                              |
| 22 | MA.6.NSO.1.4<br>easy        | >                                                                                                           | Find both absolute values, then compare those distances.                                          |
| 23 | MA.6.NSO.4.1<br>challenging | 28                                                                                                          | Adding a negative number moves left on the number line; adding a positive number moves right.     |
| 24 | MA.6.NSO.2.2<br>easy        | 18                                                                                                          | Multiply the amount for one batch by the number of batches.                                       |
| 25 | MA.6.GR.2.4<br>moderate     | $2(5 \times 5) + 2(5 \times 2) + 2(5 \times 2)$ , $2(5 \times 5 + 5 \times 2 + 5 \times 2)$ , $90 \times 1$ | Surface area adds the areas of all six faces, and opposite faces are equal.                       |
| 26 | MA.6.GR.2.2<br>moderate     | $8 \times 4$ , $4 \times 8$ , $32 \times 1$                                                                 | The area of a parallelogram is the base times the perpendicular height.                           |
| 27 | MA.6.AR.1.1<br>moderate     | $4x + 6$ , $6 + 4x$ , $x \cdot 4 + 6$                                                                       | Addition can be written in either order, but the coefficient and the constant cannot be combined. |
| 28 | MA.6.NSO.3.3<br>easy        | 0.09                                                                                                        | Squaring means multiplying the number by itself.                                                  |
| 29 | MA.6.NSO.4.2<br>challenging | -81                                                                                                         | The product of two numbers with the same sign is positive; with different signs it is negative.   |
| 30 | MA.6.AR.3.2<br>moderate     | $64 \div 8$ , $64/8$ , $128 \div 16$                                                                        | The unit rate is the total cost divided by the number of pounds.                                  |
| 31 | MA.6.NSO.2.1<br>easy        | 30.88                                                                                                       | Multiply the unit price by the number of items.                                                   |
| 32 | MA.6.AR.3.4<br>moderate     | $15\% \times 80$ , $(15/100) \times 80$ , $80 \times 0.15$                                                  | A percent of a number is that fraction of 100 multiplied by the number.                           |
| 33 | MA.6.NSO.1.1<br>easy        | -4.5, $-1 \frac{3}{4}$ , $\frac{2}{5}$ , 1                                                                  | Order the numbers by their positions on a number line, from left to right.                        |
| 34 | MA.6.NSO.4.1<br>easy        | 8                                                                                                           | Add the rise and subtract the fall from the starting temperature.                                 |
| 35 | MA.6.NSO.3.5<br>easy        | 75                                                                                                          | Multiply the decimal by 100 to write it as a percent.                                             |
| 36 | MA.6.AR.1.2<br>moderate     | 29, 34, 36                                                                                                  | Substitute each value and check whether the inequality is true.                                   |

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|----|-------------------------|------------------------------------------------------|---------------------------------------------------------------|
| 37 | MA.6.DP.1.2<br>moderate | The mean is 16., The median is 15., The range is 15. | Check each statement by computing that measure from the data. |
| 38 | MA.6.AR.2.4<br>moderate | $x + 1.5 = 7.6$ , $x - 0.5 = 5.6$ ,<br>$2x = 12.2$   | Substitute 6.1 for x and keep the equations that are true.    |